Water Surface Intelligent Floating Solar Power Generation System

CHONG NGAI HEI
Supervisor: XIAO YU HUA
Instituto Salesiano Macao
[Email address to be supplied]

Introduction / Background / Purpose

Reservoir-based floating photovoltaics offer a way to develop renewable energy while making use of existing water surfaces. This project develops an Intelligent Floating Solar Power Generation System for Macao, combining a modular floating platform with distributed sensing and wireless communication.

Methodology / Methods

Hexagonal modules were designed in Autodesk Fusion and assembled into a honeycomb arrangement. Each module integrates a solar panel, an ESP32 microcontroller, and a buck converter designed to regulate the panel's approximately 18–22 V output for a 12 V supply. Magnets and Hall-effect sensors support alignment and connection detection. ESP-NOW enables direct communication between modules and a central terminal without an external router. Prototype evaluation examined wireless range and reliability, flotation, and surface drainage. Graph-theoretic concepts were also used to guide module connection arrangements.

Results / Data / Observations

The connected modules floated stably and prevented surface water accumulation during the reported tests. Indoor communication over approximately 10 m exchanged commands and sensor data with latency of a few milliseconds and no reported packet loss across hundreds of transmissions. Outdoor links reached approximately 200–220 m under unobstructed line-of-sight conditions, although packet loss and occasional disconnections appeared beyond approximately 200 m. Multiple modules successfully shared sensor data with one another and the central terminal.

Conclusions / Applications

The prototype demonstrates the feasibility of combining modular floating solar structures with distributed monitoring. Its expandable architecture provides a basis for further development of reservoir-based renewable energy systems. Long-term durability, large-array performance, electrical conversion efficiency, and potential reductions in evaporation and panel temperature require further quantitative field evaluation.

Keywords

Floating photovoltaics; modular platform; ESP32; ESP-NOW; renewable energy.